UDAY CHARAN BOLLI

Design Verification Engineer

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PROFESSIONAL SUMMARY

Electronics & Communication Engineer with 9 months of specialized training in ASIC Design Verification from Quality Thoughts Institute, covering SystemVerilog, UVM methodology, and functional verification flows. Built strong RTL fundamentals through hands-on industry experience in semiconductor manufacturing and PCB design at Shindengen Electric and Dixon Technologies. Skilled in developing UVM-based testbenches, writing SystemVerilog assertions, and driving coverage-driven verification for digital designs. Detail-oriented and quality-focused, with a strong foundation in digital logic, Verilog HDL, and precision engineering, seeking to apply verification skills to deliver silicon-ready, bug-free designs.

TECHNICAL SKILLS

Verification: SystemVerilog, UVM (Universal Verification Methodology), Testbench Architecture, Constrained Random Verification, Functional Coverage, SystemVerilog Assertions (SVA)

HDL & Digital Design: Verilog, SystemVerilog, RTL Design, Digital Logic Design

EDA & Simulation Tools: ModelSim, EDA Playground, Xilinx Vivado, Proteus

Hardware & Embedded: Microcontrollers (AVR, ARM), Embedded C, Power Electronics, Sensors, PCB Layout & Routing

Manufacturing & Quality: SMT Assembly, Component Validation, Process Optimization, Quality Control, Failure Analysis

Debug & Test: Circuit Testing, Waveform Debugging, Multimeter, Oscilloscope

TRAINING & CERTIFICATION

Design Verification Engineer Training (9 Months)

Quality Thoughts Institute

- Completed an intensive, industry-aligned program in ASIC Design Verification covering SystemVerilog, UVM methodology, and end-to-end functional verification flows.
- Built complete UVM testbench environments (driver, monitor, sequencer, scoreboard, agent) for verifying RTL designs against protocol and functional specifications.
- Applied constrained-random stimulus generation, functional coverage, and SystemVerilog assertions to validate design correctness and close coverage gaps.
- Practiced simulation, waveform debugging, and regression analysis using industry-standard EDA simulation tools.

KEY PROJECTS (DESIGN VERIFICATION)

UVM-Based Verification of APB Protocol

- Developed a complete UVM testbench for an APB slave design, including driver, monitor, sequencer, and scoreboard components.
- Implemented constrained-random stimulus generation and functional coverage to verify protocol compliance across read/write transactions.
- Achieved high functional and code coverage through systematic test-case and corner-case development.

Functional Verification of Synchronous FIFO using SystemVerilog

- Designed a self-checking SystemVerilog testbench with a scoreboard to automatically verify FIFO read/write behavior.
- Wrote SystemVerilog assertions (SVA) to check full/empty boundary conditions and prevent overflow/underflow bugs.
- Used functional coverage to confirm all critical FIFO state transitions were exercised.

32-bit Vedic Multiplier using Carry Look-Ahead Adder

- Designed a 32-bit Vedic Multiplier based on the Urdhva-Tiryagbhyam algorithm, implemented in Verilog HDL and simulated using FPGA design tools.
- Verified functional correctness against expected outputs and benchmarked delay/speed against Array, Booth, Wallace, and Dadda multipliers.

PROFESSIONAL EXPERIENCE

Chip Mounter & PCB Design Engineer
June 2025 – September 2025

Shindengen Electric Manufacturing India (Japanese R&D Team), Bengaluru, Karnataka

- Operated and maintained precision chip mounting equipment for power supply circuit assembly in line with Japanese engineering standards.
- Conducted component validation and quality testing on power electronics PCB assemblies, strengthening attention to detail critical for verification work.
- Contributed to process improvement initiatives that reduced assembly defects and optimized production workflow.
- Designed and validated PCB layouts for power management circuits using industry-standard EDA tools.
- Interfaced with global R&D teams to implement circuit design modifications based on manufacturing feedback.

Industrial Trainee – TV Manufacturing
April 2022 – May 2022

Dixon Technologies India Limited, Renigunta, Tirupati

- Mapped the complete TV production line from component sourcing through final assembly and testing.
- Gained hands-on exposure to SMT, through-hole assembly, and quality control processes.
- Identified critical failure points where a single component error could halt production, reinforcing a precision-first engineering mindset.

EDUCATION

Bachelor of Technology (B.Tech), Electronics & Communication Engineering

Jawaharlal Nehru Technological University Anantapur (JNTUA) | CGPA: 8.4/10 | Sept 2023 – Apr 2026

Diploma in Electronics & Communications Engineering

Government Polytechnic Kalikiri | 87% Aggregate | June 2020 – May 2023

ACHIEVEMENTS

- NMMS Scholarship Recipient (National Means-cum-Merit Scholarship)
- Internship Certificate – Embedded Systems